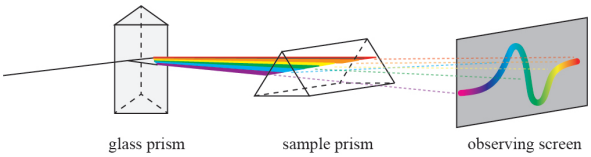
Exercises Part I

Semiclassical Quantum Optics

EXERCISE 1. CLASSICAL MODEL OF RESONANT OPTICAL RESPONSE

In 1902, Robert Williams Wood expanded on August Kundt's method of crossed prisms to map out the refractive index of sodium vapor near its famous "D" absorption line. In a classic crossed-prisms experiment, white light passes through a vertical glass prism, which spreads the colors out horizontally into a spectrum (like a standard rainbow). This horizontally dispersed light then passes through a horizontal prism filled with sodium vapor.



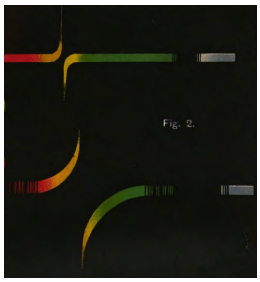


FIG. 1. Crossed-prisms experiment and the observed spectrum for Exercise 1. The two original image files must be placed at the paths indicated if they are not already present.

(a) Model the optical susceptibility using the Lorentz model. Determine the refractive index and absorption coefficient assuming low absorption.

(b) Comment and explain the observations of the Fig. 1.

(c) Make a model for the deflection observed on the screen using Snell's law, and comment on how to determine the free parameters of the Lorentz model from these observations.

(d) Can you observe slow light (i.e. ultra-low light velocities) in this experiment? How would you do it?

EXERCISE 2. THE MOLLWO-IVEY RELATION

An F centre consists of an electron trapped at a missing anion in an ionic crystal. We may approximate the electron as a particle of mass m_e confined to a three-dimensional cubic

infinite potential well of side length a:

$$V(x,y,z) = \begin{cases} 0, & 0 < x, y, z < a, \\ +\infty, & \text{otherwise.} \end{cases} \tag{1}$$

Assume that the confinement length a is proportional to the crystal lattice spacing, and neglect electron spin.

(a) Solve the time-independent Schrödinger equation by separation of variables. Apply the boundary conditions and normalize the wavefunctions. Show that the energy eigenvalues are

$$E_{n_x n_y n_z} = \frac{\pi^2 \hbar^2}{2 m_e a^2} \left(n_x^2 + n_y^2 + n_z^2 \right), \qquad n_x, n_y, n_z = 1, 2, \ldots . \tag{2}$$

(b) Identify the ground state and the first excited energy level. Calculate their energies and determine the degeneracy of the first excited level.

(c) For light polarized along x, the dipole interaction is $\hat{V}(t) = eE(t)\hat{x}$, where $e > 0$. Using the product structure of the wavefunctions, determine which states of the first excited level can be reached from the ground state by an electric-dipole transition. Verify that the relevant matrix element of $\hat{x}$ is nonzero.

(d) Let ν_{abs} denote the frequency of the lowest allowed absorption transition. Use

$$h\nu_{\text{abs}} = E_{\text{exc}} - E_{\text{g}} \tag{3}$$

to derive the dependence of ν_{abs} on a. Show that

$$\nu_{\text{abs}} a^2 = \text{constant}, \tag{4}$$

and calculate the constant within this model. This inverse-square scaling is the particle-in-a-box prediction for the *Mollwo–Ivey relation*.

(e) For two crystals described by the same model, show that

$$\ln\!\left(\frac{a_2}{a_1}\right) = -\frac{1}{2} \ln\!\left(\frac{\nu_{\text{abs},2}}{\nu_{\text{abs},1}}\right). \tag{5}$$

Does increasing the lattice spacing produce a red shift or a blue shift? Calculate the percentage change in the absorption frequency when the confinement length increases by 10%. Finally, discuss two limitations of modelling a real colour centre as an infinite cubic well.

EXERCISE 3. RABI OSCILLATIONS, POPULATION LOSS, AND THE DENSITY MATRIX

Consider a two-level atom driven by a classical field $E(t) = E_0 \cos \omega t$, with Hamiltonian

$$\hat{H}(t) = \hbar\omega_0 |2\rangle\langle 2| - \hbar\Omega \cos \omega t \left(|2\rangle\langle 1| + |1\rangle\langle 2| \right), \tag{6}$$

where $\Omega = dE_0/\hbar$ is real and $\Delta = \omega_0 - \omega$. The atom is initially excited: $|\psi(0)\rangle = |2\rangle$.



FIG. 2. Level schemes for Exercise 3. (a) Original diagram from Rand, Exercise 3.10 (p. 80), as in the supplied screenshot: both levels decay to states outside the retained manifold. The book's γ_1, γ_2 correspond to Γ_1, Γ_2 below. (b) Companion schematic for the closed manifold discussed in Exercise 3.11: the drive couples the two states and spontaneous emission returns population from $|2\rangle$ to $|1\rangle$.

(a) Starting from the Schrödinger equation, transform to the interaction picture with respect to $\hat{H}_0 = \hbar\omega_0 |2\rangle\langle 2|$. Identify the terms oscillating at $\omega_0 - \omega$ and $\omega_0 + \omega$, and state the conditions for the rotating-wave approximation (RWA).

After applying the RWA, introduce rotating-frame amplitudes through

$$C_1 = e^{-i\Delta t/2} b_1, \qquad C_2 = e^{i\Delta t/2} b_2. \tag{7}$$

Derive the resulting time-independent Hamiltonian and solve for the populations $P_1(t)$ and $P_2(t)$. Identify the generalized Rabi frequency, discuss the resonant limit, and verify that $P_1 + P_2 = 1$.

(b) Now suppose that population leaves each level for states outside the two-level manifold [Fig. 2(a)], at rates Γ_1 and Γ_2, respectively. The interaction-picture amplitudes then

satisfy

$$\begin{aligned}\dot{C}_2 &= -\frac{\Gamma_2}{2}C_2 + \frac{i\Omega}{2}e^{i\Delta t}C_1,\\ \dot{C}_1 &= -\frac{\Gamma_1}{2}C_1 + \frac{i\Omega}{2}e^{-i\Delta t}C_2.\end{aligned} \tag{8}$$

Use the same rotating-frame transformation to solve these equations for $C_2(0) = 1$ and $C_1(0) = 0$, and obtain $|C_1(t)|^2$ and $|C_2(t)|^2$. Show that

$$\frac{d}{dt}\left(|C_1|^2 + |C_2|^2\right) = -\Gamma_1|C_1|^2 - \Gamma_2|C_2|^2. \tag{9}$$

Interpret the missing probability and recover part (a) when both decay rates vanish.

(c) For a closed two-level manifold, spontaneous emission returns population from $|2\rangle$ to $|1\rangle$ [Fig. 2(b)]. Consider the proposed amplitude equations

$$\begin{aligned}\dot{C}_2 &= -\frac{\Gamma}{2}C_2 + \frac{i\Omega}{2}e^{i\Delta t}C_1,\\ \dot{C}_1 &= +\frac{\Gamma}{2}C_2 + \frac{i\Omega}{2}e^{-i\Delta t}C_2.\end{aligned} \tag{10}$$

Calculate the time derivative of $|C_1|^2 + |C_2|^2$ and show that these equations do not generally preserve normalization. Evaluate that derivative at $t = 0$ for the specified initial state.

Explain why transferring population at rate $\Gamma|C_2|^2$ is not equivalent to adding an amplitude source $\Gamma C_2/2$. Write instead the spontaneous-emission contribution to $\dot{\rho}_{11}$, $\dot{\rho}_{22}$, and $\dot{\rho}_{12}$, and verify that the density-matrix description preserves $\rho_{11} + \rho_{22} = 1$.